CMR Technical Campus
Department of Information Technology

Major Project Stage – II
Review- 2

Project Title:Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

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INTRODUCTION AND OVERVIEW

- Brand name is very important for any business because it creates first impression.
- A good brand name helps in marketing, recognition, and customer attraction.
- Traditional methods like brainstorming take more time and effort.
- These methods also depend on human creativity and may not give best results.
- Our project uses AI and NLP to automate this process.
- It helps in generating and evaluating brand names efficiently

EXISTING SYSTEM

- Brand names are created manually by marketing teams.
- Teams use brainstorming sessions to generate ideas.
- Some tools generate names using keywords.
- Rule-based methods are also used for name generation.
- These systems do not understand meaning or context of words.
- There is no proper evaluation of generated names.

DISADVANTAGES OF EXISTING SYSTEM

- It takes a lot of time to generate names.
- Results are subjective and depend on individual ideas.
- No proper system to evaluate quality of names.
- Cannot handle large data or multiple ideas.
- Generated names may be repetitive or not unique.
- Does not consider meaning, emotion, or market trends.

PROPOSED SYSTEM

- Our system uses Artificial Intelligence and NLP techniques.
- First, text data is cleaned using preprocessing methods.
- Then we use DistilRoBERTa model to understand context of words.
- This model converts text into meaningful embeddings.
- After that, machine learning models are applied:
- Logistic Regression
- Random Forest
- Support Vector Machine
- We use SMOTE to balance dataset.
- Finally, results are shown using Flask web application.

ADVANTAGES OF PROPOSED SYSTEM

- Saves time compared to manual methods.
- Provides more accurate and reliable results.
- Reduces human effort and errors.
- Can handle large datasets efficiently.
- Uses data-driven approach instead of guessing.
- Improves creativity using AI suggestions.
- Gives better decision support for branding.

SOFTWARE & HARDWARE REQUIREMENTS

SOFTWARE REQUIRMENTS :

Operating System	:	Windows
Backend Technologies	:	Python 3.12.0
Framework	:	Flask
Database	:	MySQL

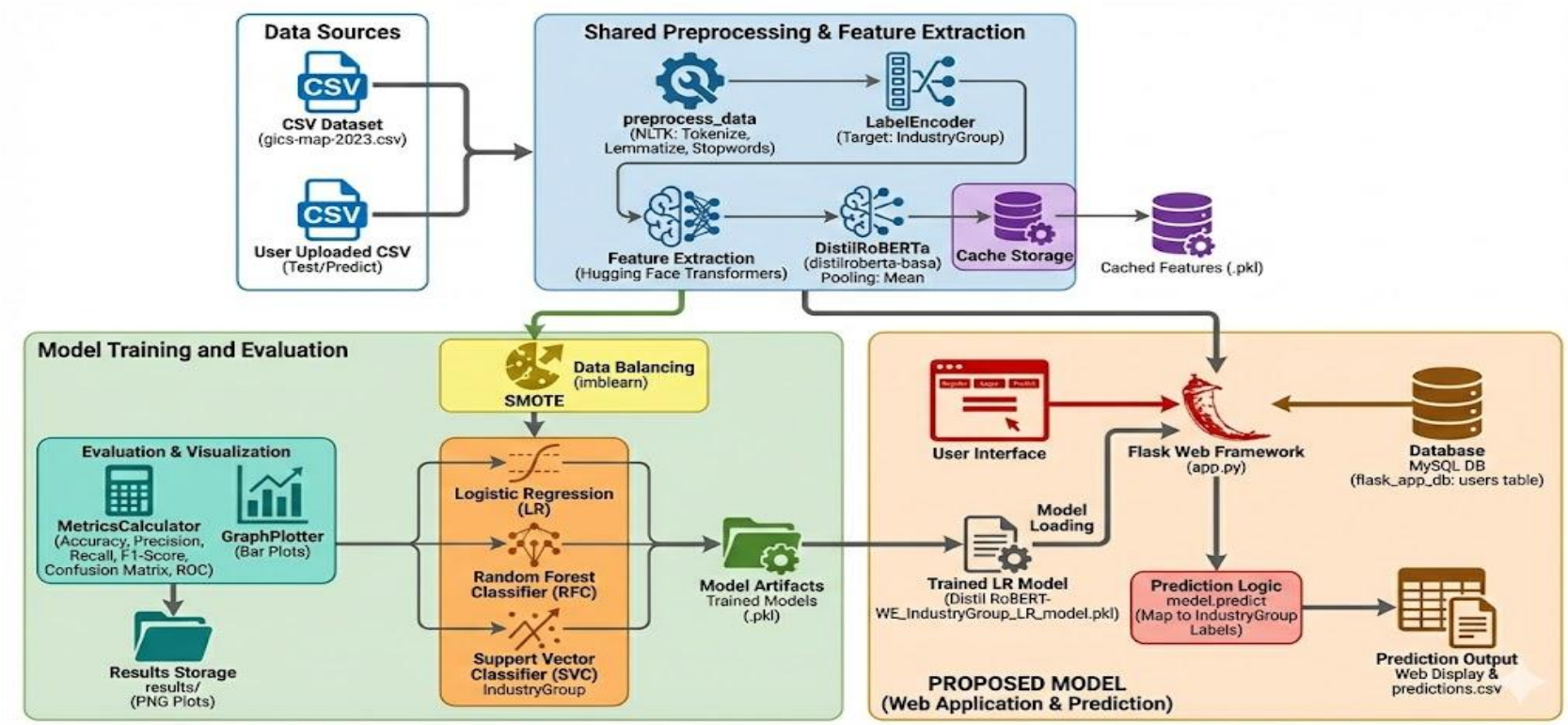
HARDWARE REQUIRMENTS :

Processor	:	Intel i3 and above
Hard Disk	:	Minimum 512GB
RAM	:	Minimun 8GB

NOVELTY OF THE PROJECT

- This project uses Artificial Intelligence and NLP instead of manual methods.
- It uses DistilRoBERTa, a transformer model, to understand the meaning and context of text, not just keywords.
- The system converts text into semantic embeddings for better analysis.
- It combines Machine Learning models (Logistic Regression, Random Forest, SVM) for accurate prediction.
- It uses SMOTE technique to balance the dataset and improve model performance.
- The system provides automatic evaluation of brand names instead of human guessing.
- It is integrated with a Flask web application for real-time user interaction.
- The system can handle large datasets efficiently and gives fast results.

SYSTEM ARCHITECTURE



MODULES

- Data Input Module
- Preprocessing Module
- NLP Processing Module
- Embedding Generation Module
- Data Balancing Module (SMOTE)
- Model Training Module
- Evaluation Module
- Prediction Module
- Visualization Module
- Web Interface Module

Data Input Module

- This module allows users to provide input data in the form of text or CSV files through the web interface.
- It ensures that the input data is properly received and forwarded to the preprocessing stage.

Preprocessing Module

- This module cleans the raw text by removing noise, special characters, and converting text into lowercase.
- It also performs tokenization, stopword removal, and lemmatization to make the data ready for analysis.

NLP Processing Module

•This module applies Natural Language Processing techniques to understand the structure and meaning of the text.

•It helps in identifying important words and patterns required for accurate prediction.

Embedding Generation Module

•This module uses the DistilRoBERTa model to convert text into numerical vectors (embeddings).

•These embeddings capture the semantic meaning and context of the text for machine learning models.

Data Balancing Module (SMOTE)

- This module handles class imbalance using SMOTE technique.
- It generates synthetic data for minority classes to improve model performance.

Model Training Module

•This module trains machine learning models such as Logistic Regression, Random Forest, and SVM.

•It uses balanced data (with SMOTE) to improve accuracy and performance of the models.

Evaluation Module

•This module evaluates model performance using metrics like accuracy, precision, recall, and F1-score.

•It helps in comparing different models and selecting the best-performing model.

Prediction Module

•This module takes new input data and uses the trained model to predict the output.

•It generates the final result such as category or evaluation of the brand name.

Visualization Module

•This module creates graphs like word clouds, frequency charts, and confusion matrix.

•It helps in understanding data patterns and model performance visually.

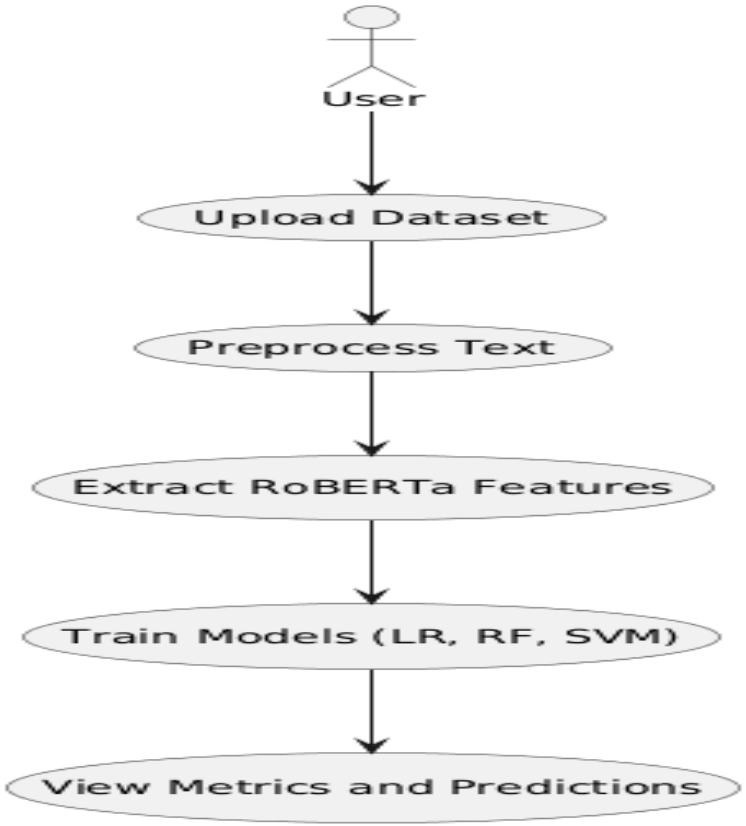
Web Interface Module

•This module is developed using Flask with HTML and CSS for user interaction.

•It displays the prediction results, metrics, and allows users to easily interact with the system.

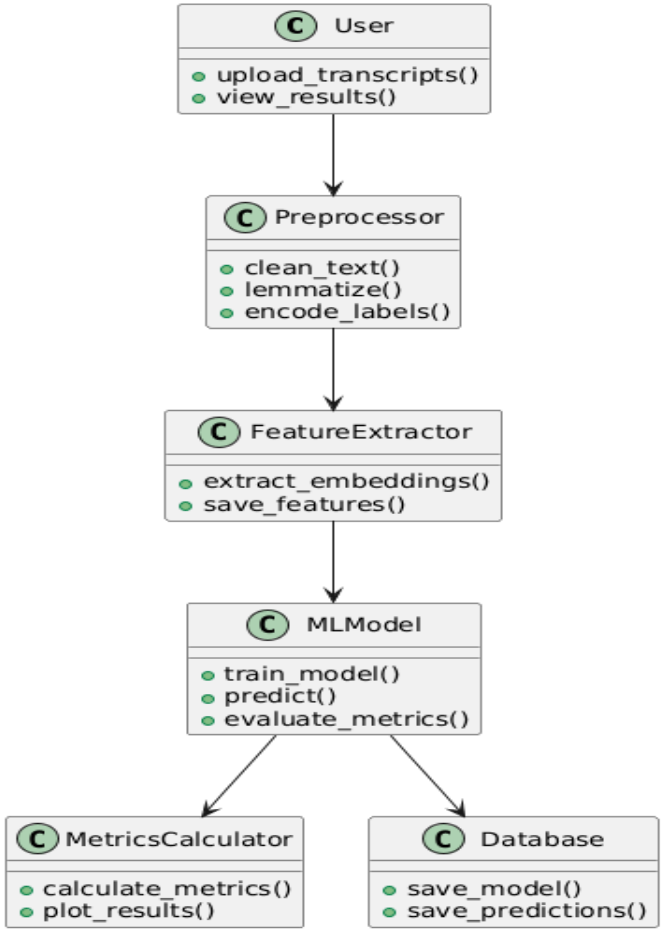
UML DIAGRAMS

Use Case Diagram



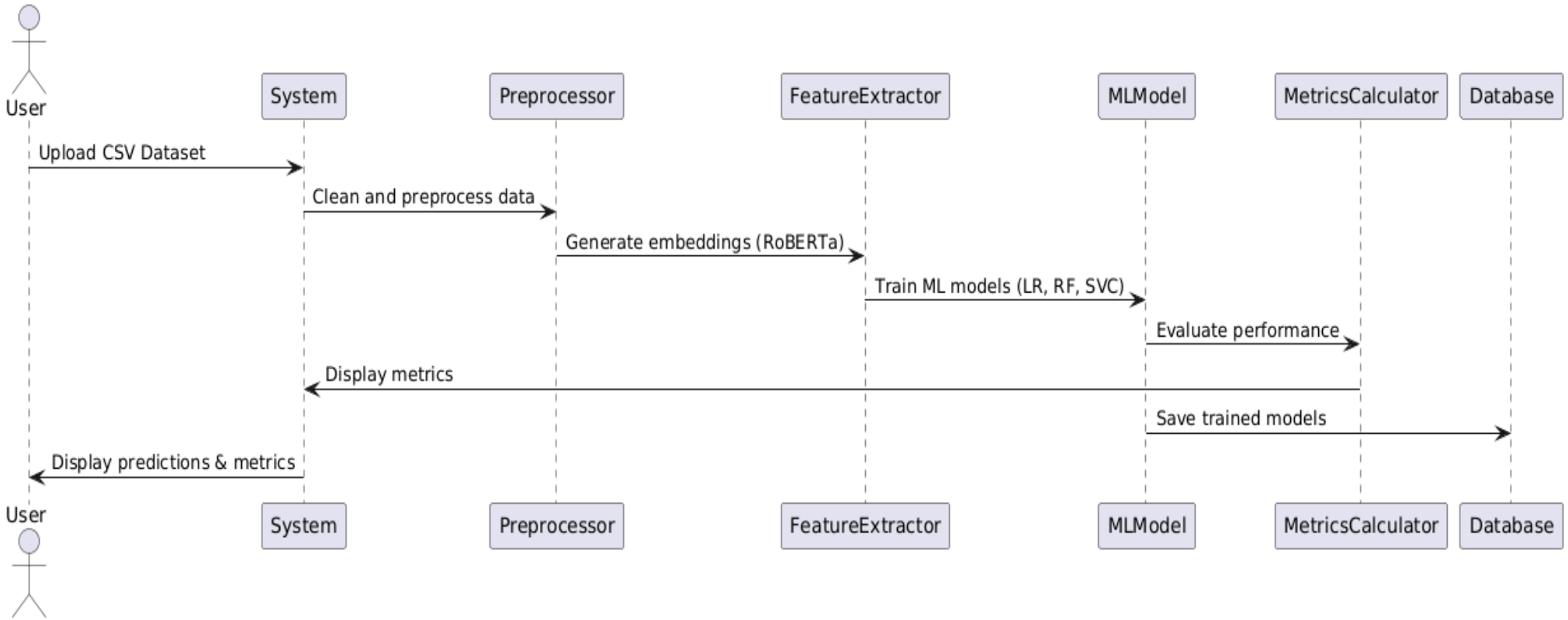
UML DIAGRAMS

Class Diagram



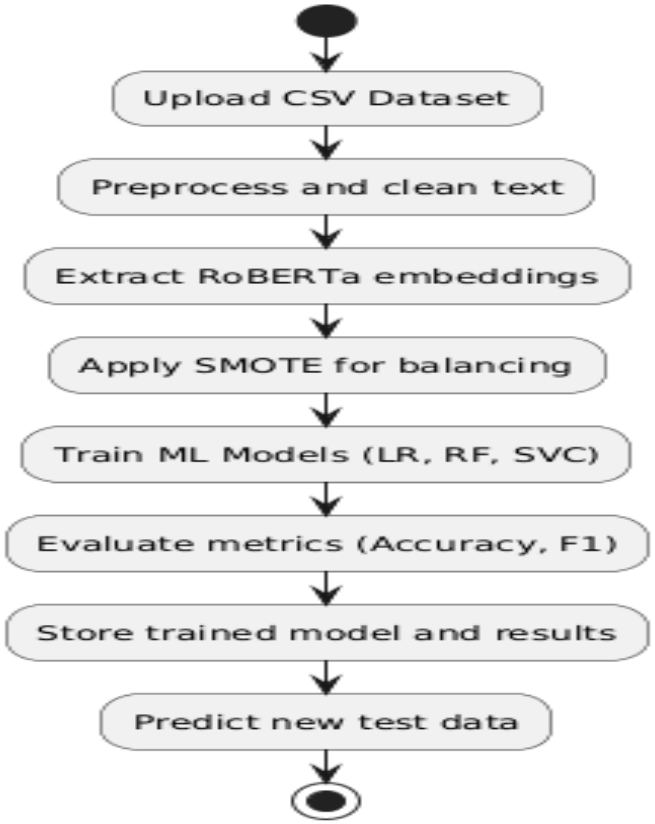
UML DIAGRAMS

Sequence Diagram



UML DIAGRAMS

Activity Diagram



SOURCE CODE

PREPROCESSING :

```
def preprocess_text_data(df):
    if 'text' in df.columns:
        df['clean_text'] = df['text'].str.lower().str.replace(r'[^a-z\s]', '', regex=True)
    else:
        df['clean_text'] = df.iloc[:, 0].astype(str).str.lower()
    return df
```

MODEL LOADING :

```
def load_model(model_name):
    model_path = os.path.join("model", model_name)
    return joblib.load(model_path)
```

SOURCE CODE

PREDICTION :

```
df = pd.read_csv(file)
df = preprocess_text_data(df)
model = load_model("Distil_RoBERT-WE_IndustryGroup_LR_model.pkl")
features = df['clean_text'].apply(lambda x: len(x.split())).values.reshape(-1, 1)
y_pred = model.predict(features)
df['Predicted_Label'] = y_pred
metrics = {
    "Accuracy": accuracy_score(y_true, y_pred),
    "Precision": precision_score(y_true, y_pred, average="weighted"),
    "Recall": recall_score(y_true, y_pred, average="weighted"),
    "F1-Score": f1_score(y_true, y_pred, average="weighted")
}
```

SOURCE CODE

EVALUATION :

```
@app.route('/predict', methods=['GET', 'POST'])
def predict():
    file = request.files.get('file')
    df = pd.read_csv(file)
    df = preprocess_text_data(df)
    return render_template('predict.html')
```

RESULTS

System Home Page

- Main interface of the system
- Provides navigation options
- User-friendly design

Fig 1 : System Home Page

RESULTS

Registration Page

- New user account creation
- Collects user details
- Stores data in database

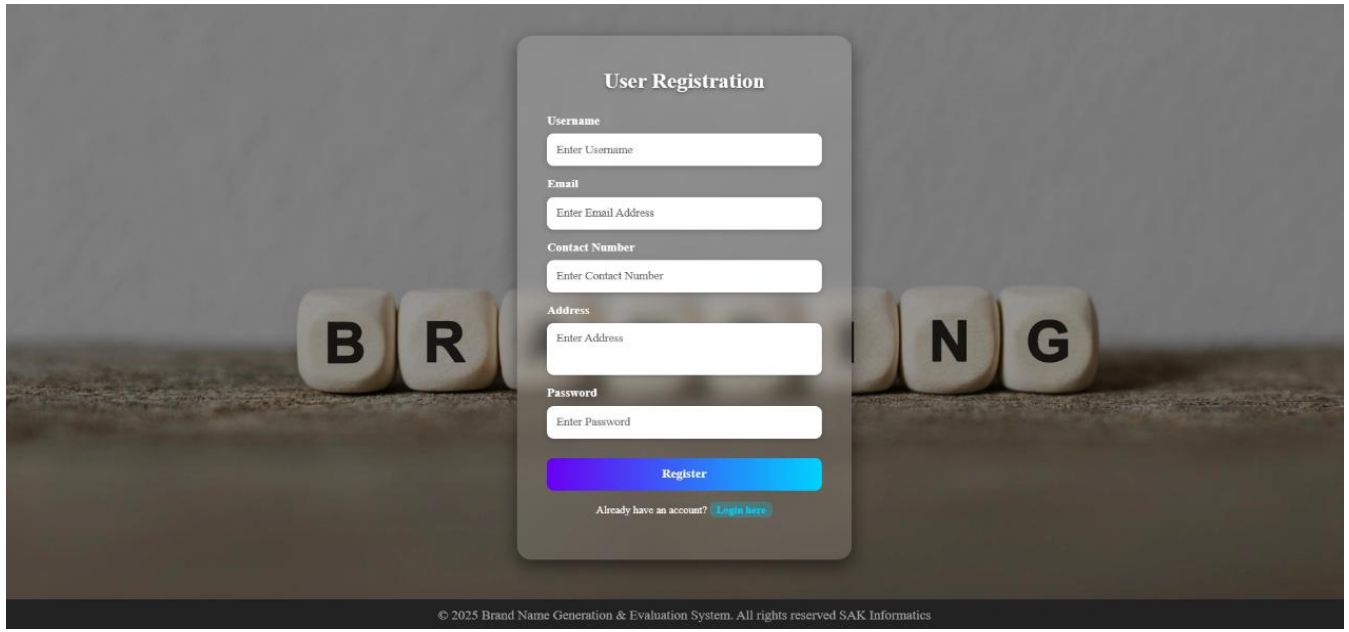


Fig 2 : Registration Page

RESULTS

Login Page

- User authentication
- Secure access to system
- Prevents unauthorized entry

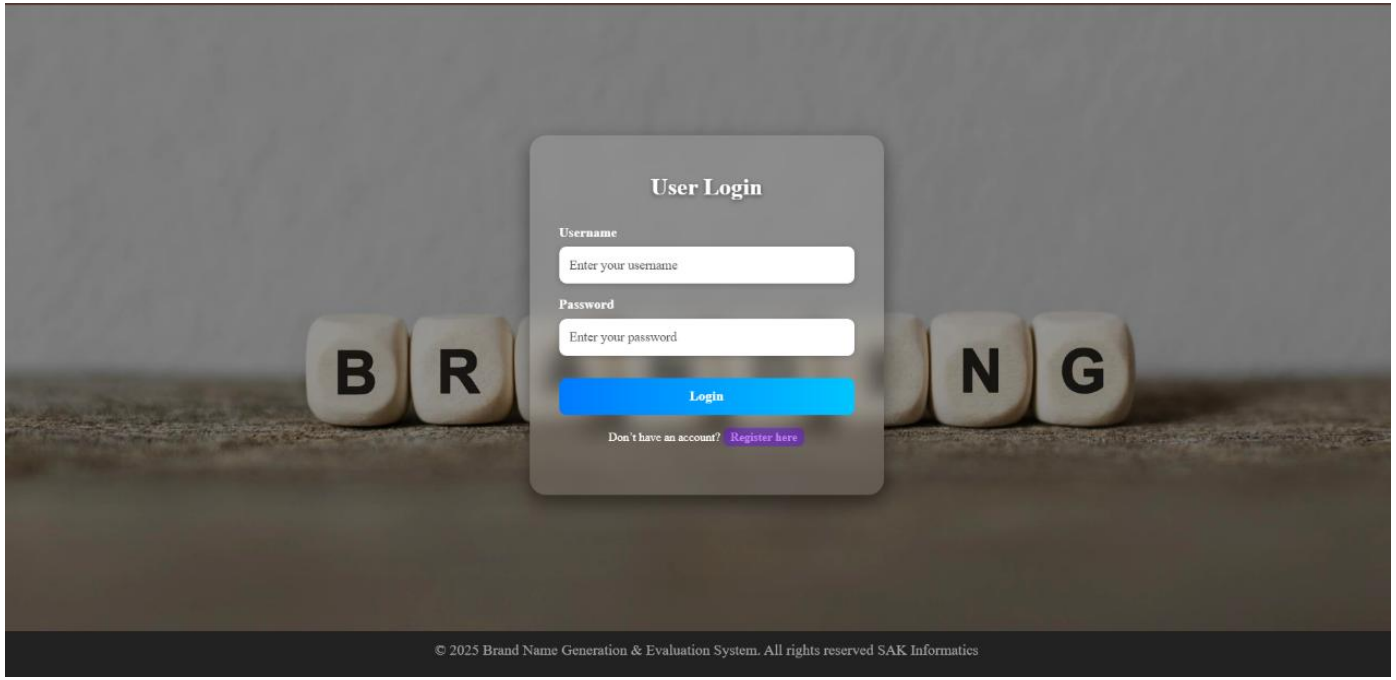


Fig 3 : Login Page

RESULTS

Uploading Test Dataset Page

- Upload CSV file
- Validates input data
- Sends data for prediction

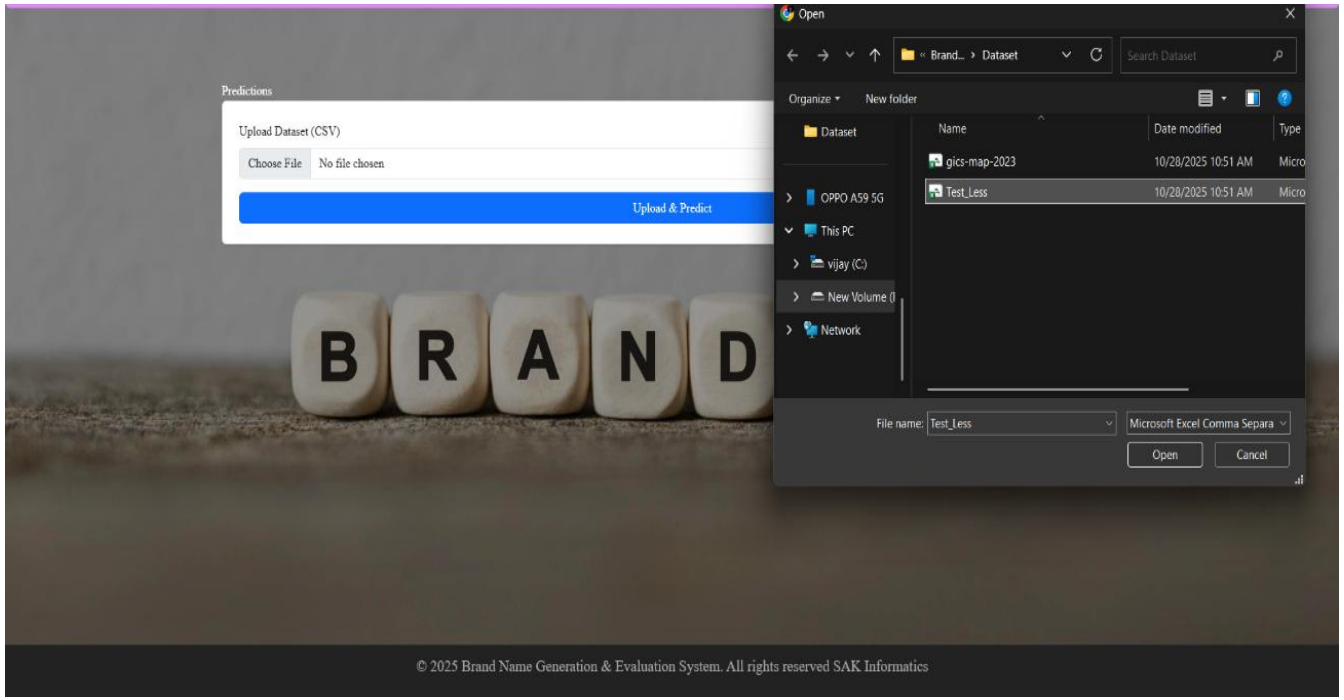


Fig 4 : Uploading Test Dataset

RESULTS

Prediction Results Page

- Displays predicted output
- Shows results in table format
- Provides model results

Choose File | No file chosen

Upload & Predict

Prediction Results

SectorId	Sector	IndustryGroupId	IndustryId	Industry	SubIndustryId	SubIndustry	SubIndustryDescription	Predicted_IndustryGroup
10	Energy	1010	101010	Energy Equipment & Services	10101010	Oil & Gas Drilling	Drilling contractors or owners of drilling rigs that contract their services for drilling wells.	Energy
10	Energy	1010	101010	Energy Equipment & Services	10101020	Oil & Gas Equipment & Services	Manufacturers of equipment, including drilling rigs and equipment, and providers of supplies and services to companies involved in the drilling, evaluation and completion of oil and gas wells.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102010	Integrated Oil & Gas	Integrated oil companies engaged in the exploration & production of oil and gas, as well as at least one other significant activity in either refining, marketing and transportation, or chemicals.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102020	Oil & Gas Exploration & Production	Companies engaged in the exploration and production of oil and gas not classified elsewhere.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102030	Oil & Gas Refining & Marketing	Companies engaged in the refining and marketing of oil, gas and/or refined products not classified in the Integrated Oil & Gas or Independent Power Producers & Energy Traders Sub-Industries.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102040	Oil & Gas Storage & Transportation	Companies engaged in the storage and/or transportation of oil, gas and/or refined products. Includes diversified midstream natural gas companies, oil and refined product pipelines, coal slurry pipelines and	Energy

Fig 5 : Prediction Results Page

CONCLUSION & FUTURE ENHANCEMENT

Future Enhancement

- Improve accuracy using better models
- Improve user interface
- Support multiple languages
- Add real-time data analysis
- Generate brand names automatically

Conclusion

- AI and NLP improve brand name generation.
- System is fast, efficient, and accurate.
- Reduces manual effort and errors.

KEY REFERENCES

- Maria DSouza Deryl, Sanjeev Verma, Vartika Srivastava,How does AI drive branding? Towards an integrated theoretical framework for AI-driven branding,International Journal of Information Management Data Insights, Volume 3, Issue 2,2023,100205, ISSN 2667-0968, https://doi.org/10.1016/j.jjimei.2023.100205.

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- Gerlich, M. The Shifting Influence: Comparing AI Tools and Human Influencers in Consumer Decision-Making. AI **2025**, 6, 11. https://doi.org/10.3390/ai6010011

GITHUB LINK

Link → https://github.com/PrachetanaGodishala/NLP-Brand-Name-Industry-Classification

Thank You